

IN5340/IN9340 Project III

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Code: `github.com/Twaal/Statistical_Signal_Processing_IN5340/tree/master/Project_3`

Agenda

1 Introduction

2 Main results

Problem statement

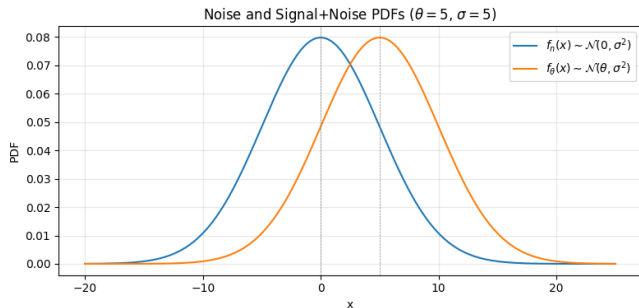
- Binary detection with additive Gaussian noise:

$$H_0 : x = n, \quad H_1 : x = \theta + n, \quad n \sim \mathcal{N}(0, \sigma^2)$$

- Use $\theta = 5$ and $\sigma = 5$ as baseline case.
- Design and evaluate:
 - CFAR detector for given P_{FA}
 - ROC curve (theoretical vs empirical)
 - Improvement using sample mean of N measurements

b) Noise and signal+noise PDFs

- Under H_0 : $x \sim \mathcal{N}(0, \sigma^2)$
- Under H_1 : $x \sim \mathcal{N}(\theta, \sigma^2)$
- For $\theta/\sigma = 1$, distributions overlap strongly.
- Detection is possible, but not highly reliable in this regime.

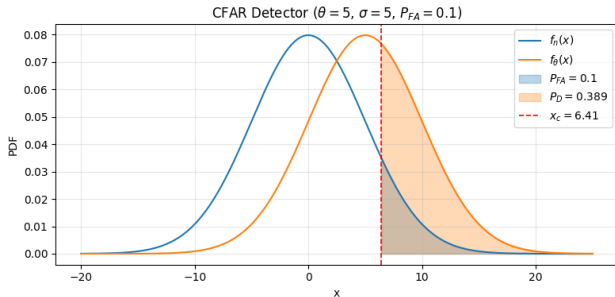


c) CFAR detector with $P_{FA} = 0.1$

$$x_c = \sigma Q^{-1}(P_{FA}) = 5 Q^{-1}(0.1) \approx 6.41$$

$$P_D = Q\left(\frac{x_c - \theta}{\sigma}\right) \approx 0.389$$

- Blue shaded area: false-alarm probability P_{FA} .
- Orange shaded area: detection probability P_D .
- Dashed red line: threshold x_c .

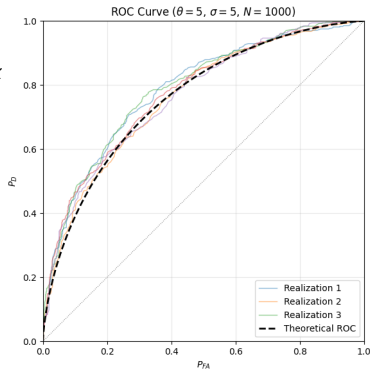


d) ROC: empirical vs theoretical

- Data model:

$$s \in \{0, \theta\}, P(s = 0) = P(s = \theta) = 0.5, x$$

- Empirical ROC computed by threshold sweep on simulated data.
- Multiple realizations vary, but cluster around the theoretical ROC.



e) Multi-sample detector and minimum N

Using sample mean of N measurements:

$$\bar{x}|H_0 \sim \mathcal{N}\left(0, \frac{\sigma^2}{N}\right), \quad \bar{x}|H_1 \sim \mathcal{N}\left(\theta, \frac{\sigma^2}{N}\right)$$

For CFAR with requirements $P_{FA} = 0.01$ and $P_D > 0.9$:

$$N \geq \left(\frac{\sigma}{\theta}\right)^2 (Q^{-1}(P_{FA}) + Q^{-1}(1 - P_D))^2$$

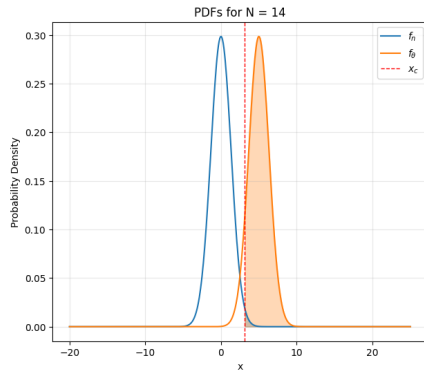
Numerically:

$$N_{\min, \text{cont}} \approx 13.02 \quad \Rightarrow \quad N_{\min} = 14$$

At $N = 14$: $x_c \approx 3.11$ and $P_D \approx 0.922$.

f) PDFs for improved detector ($N = 14$)

- Variance shrinks from σ^2 to σ^2/N .
- The two PDFs become better separated.
- This improves detection performance for fixed P_{FA} .

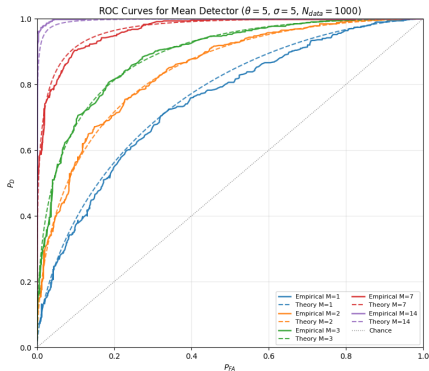


g) ROC for multiple measurements

From stdout (single realization):

- $M = 1$: $AUC_{emp} = 0.745$, $AUC_{th} = 0.760$
- $M = 2$: $AUC_{emp} = 0.843$, $AUC_{th} = 0.841$
- $M = 3$: $AUC_{emp} = 0.888$, $AUC_{th} = 0.890$
- $M = 7$: $AUC_{emp} = 0.965$, $AUC_{th} = 0.969$
- $M = 14$: $AUC_{emp} = 0.998$, $AUC_{th} = 0.996$

ROC shifts toward top-left as M increases.

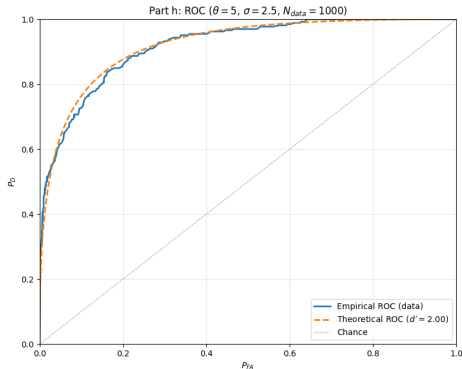


h) Imbalanced prior data ($\theta = 5$, $\sigma = 2.5$)

From stdout:

- Generated prevalence: $\pi_1 = 0.266$, $\pi_0 = 0.734$
- $d' = \theta/\sigma = 2.000$
- $AUC_{emp} = 0.918$, $AUC_{th} = 0.921$

Empirical ROC still matches theory well.

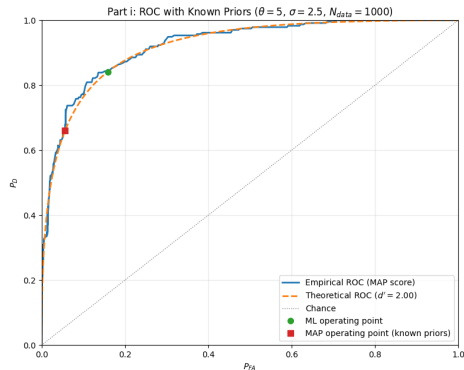


i) Known priors (MAP operating point)

From stdout:

- ML threshold: $x_t = 2.500$
- MAP threshold: $x_t = 3.968$
- ML point: $(P_{FA}, P_D) = (0.159, 0.841)$
- MAP point: $(P_{FA}, P_D) = (0.056, 0.660)$

Benefit of MAP: fewer false alarms when H_1 is rare.

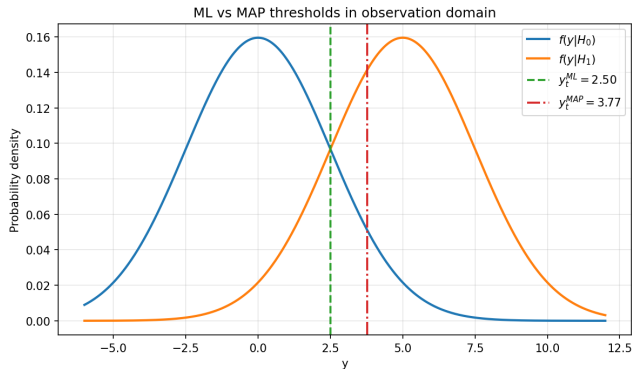


i) PDFs with ML and MAP thresholds

Interpretation:

- ML ignores class priors and uses midpoint threshold.
- MAP accounts for rare H_1 and shifts threshold right.
- Shift reduces false alarms, but also lowers detection probability.

$$x_t^{ML} = \frac{\theta}{2}, \quad x_t^{MAP} = \frac{\theta}{2} + \frac{\sigma^2}{\theta} \log \frac{P_0}{P_1}$$



MAP derivation from likelihoods

For Gaussian shift model ($y|H_0 \sim \mathcal{N}(0, \sigma^2)$, $y|H_1 \sim \mathcal{N}(\theta, \sigma^2)$):

$$f_0(y) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{y^2}{2\sigma^2}}, \quad f_1(y) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y-\theta)^2}{2\sigma^2}}$$

$$\log \frac{f_1(y)}{f_0(y)} = -\frac{(y-\theta)^2}{2\sigma^2} + \frac{y^2}{2\sigma^2} = \frac{\theta}{\sigma^2}y - \frac{\theta^2}{2\sigma^2}$$

MAP rule (equal costs):

$$\begin{aligned} \Pr(H_1 | y) &\underset{H_0}{\overset{H_1}{\geq}} \Pr(H_0 | y) \\ \frac{f_1(y)}{f_0(y)} &\underset{H_0}{\overset{H_1}{\geq}} \frac{\Pr(H_0)}{\Pr(H_1)} = \frac{P_0}{P_1} \\ \Rightarrow y &\underset{H_0}{\overset{H_1}{\geq}} y_t^{MAP} = \frac{\theta}{2} + \frac{\sigma^2}{\theta} \log \frac{P_0}{P_1} \end{aligned}$$

How priors change the detector

- Priors enter only through the term $\log(P_0/P_1)$ in the MAP threshold.
- If H_1 is rare ($P_1 < P_0$), then $\log(P_0/P_1) > 0$ and threshold increases.
- Higher threshold means fewer false alarms, but also fewer detections.

ML is a special case of MAP:

$$P_0 = P_1 \Rightarrow \log \frac{P_0}{P_1} = 0 \Rightarrow y_t^{MAP} = \frac{\theta}{2} = y_t^{ML}$$

So ML corresponds to assuming equal priors; MAP generalizes this when prior probabilities are known.

1) Multi-measurement ROC model (g):

```
d_prime_n = theta * np.sqrt(M) / sigma  
pd_theory = norm.sf(norm.isf(pfa_grid) - d_prime_n)
```

2) MAP score with known priors (i):

```
llr = (theta/sigma**2)*y - theta**2/(2*sigma**2)  
score_map = llr + np.log(P1/P0)
```

3) MAP threshold in observation domain:

$$y_t^{MAP} = \frac{\theta}{2} + \frac{\sigma^2}{\theta} \log \frac{P_0}{P_1}$$

Summary

- For single measurement ($\theta = \sigma = 5$), overlap is substantial and P_D is modest at practical P_{FA} .
- CFAR with $P_{FA} = 0.1$ gives threshold $x_c \approx 6.41$ and $P_D \approx 0.389$.
- ROC simulations agree with theory in all experiments (typical AUC match within a few hundredths).
- Averaging measurements strongly improves performance (AUC from 0.745 at $M = 1$ to 0.998 at $M = 14$).
- With imbalanced priors, MAP shifts operating point to reduce false alarms ($0.159 \rightarrow 0.066$) at the cost of lower P_D .